Task-22:-Technical Documentation(Setting Up a GitLab CE Server and Pushing a Flask Hello World Application code)

# 1. Introduction

### 1.1 Purpose

The purpose of this project is to deploy a self-hosted GitLab Community Edition (CE) server on AWS EC2 and use it as a version control platform to host a Flask Hello World application repository. This project demonstrates internal code hosting, repository management, and Git operations using a self-managed GitLab server.

### 1.2 Scope

This project includes:

- GitLab CE installation on EC2
- GitLab web interface configuration
- Repository creation in GitLab
- Flask Hello World application creation
- Pushing code to GitLab repository
- Validation of Git operations

This project does not include:

- CI/CD pipeline setup
- GitLab Runner configuration
- Production deployment

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## 2. Project Overview

### 2.1 Objective

The objective of this project is to:

- Install GitLab CE on AWS EC2
- Access GitLab via web browser
- Create a repository in GitLab
- Push Flask application code to GitLab
- Verify repository and code upload

### 2.2 Tools and Technologies Used

| Tool | Purpose |
|---|---|
| AWS EC2 | Hosting GitLab Server |
| GitLab CE | Version Control System |
| Git | Source Code Management |
| Flask | Python Web Framework |
| Ubuntu | Operating System |

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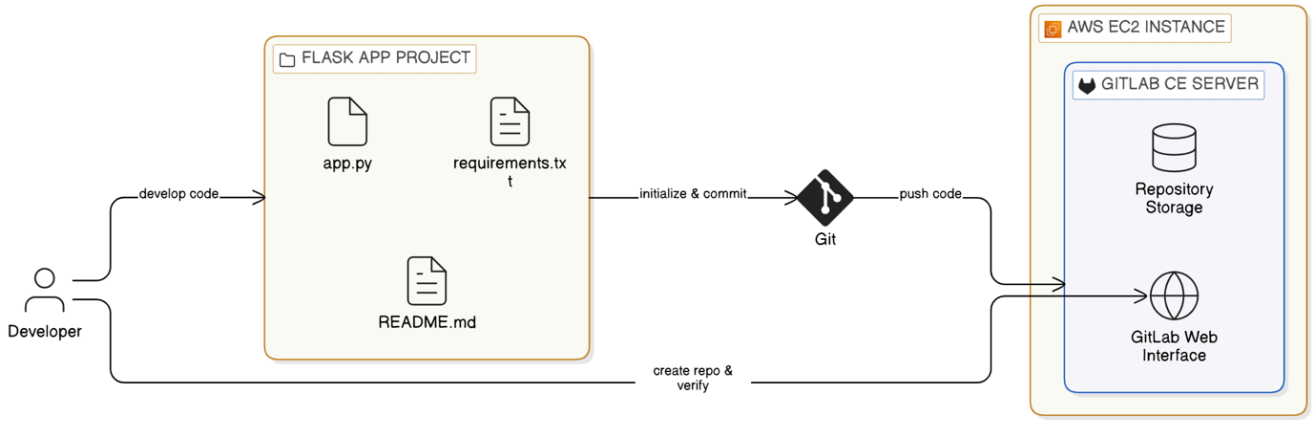
## 3. Architecture Overview

**Architecture Flow**

```
Developer → Git → GitLab CE Server → Repository Storage
```

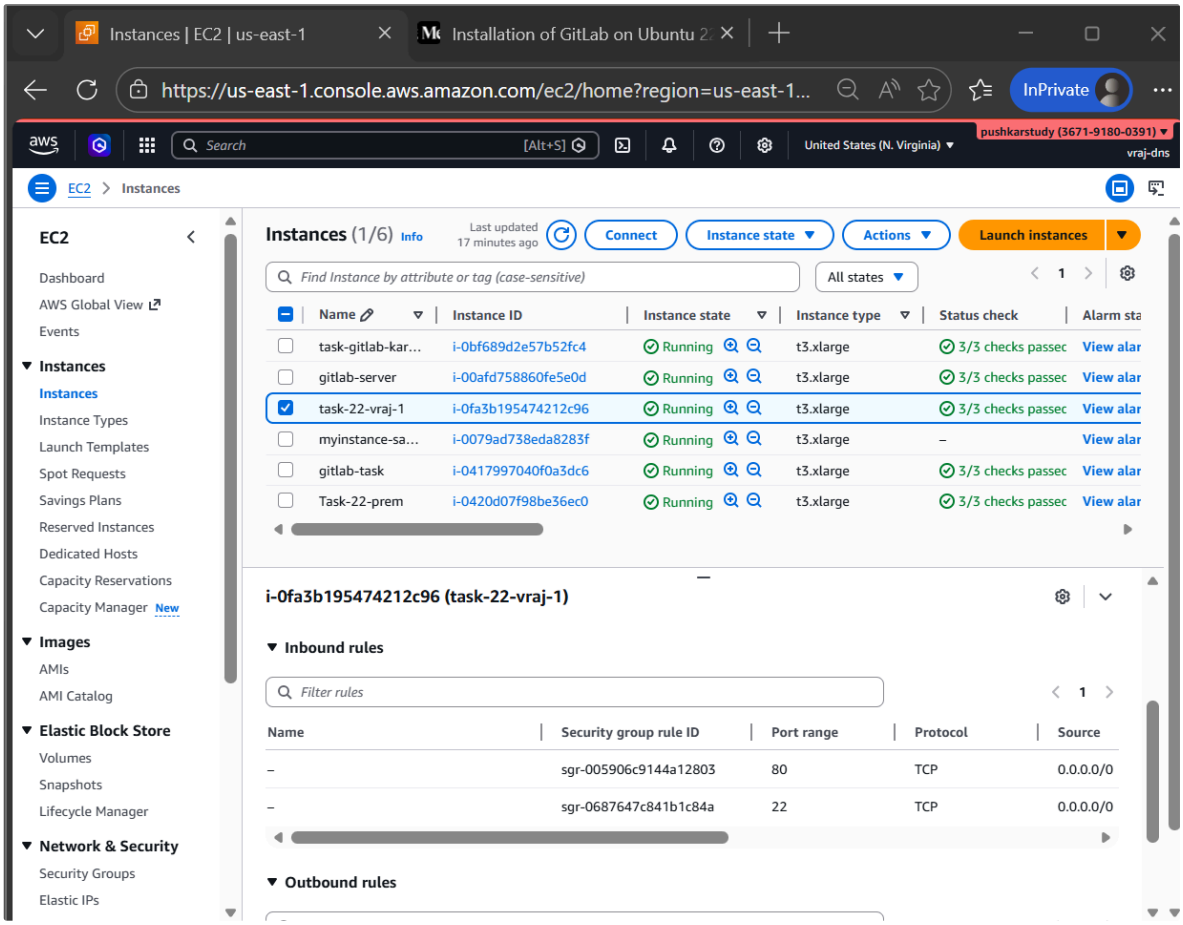



## 4. GitLab CE Server Setup

### 4.1 EC2 Instance Setup

An EC2 instance was launched with the following configuration:

| Parameter | Value |
|---|---|
| OS | Ubuntu |
| Instance Type | t2.medium |
| Storage | 20 GB |
| Open Ports | 22, 80, 443 |



### 4.2 Install GitLab CE

Commands used:

```
sudo apt update
sudo apt install -y curl openssh-server ca-certificates tzdata perl
curl https://packages.gitlab.com/install/repositories/gitlab/gitlab-ce/script.deb.sh | sudo bash
sudo apt install gitlab-ce -y
```

After installation, GitLab was configured using:

```
sudo gitlab-ctl reconfigure
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-41-179:~$ ls
ubuntu@ip-172-31-41-179:~$ cd /tmp
ubuntu@ip-172-31-41-179:/tmp$ curl -LO https://packages.gitlab.com/install/repositories/gitlab/gitlab-ce/script.deb.sh
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  5291  100  5291    0     0  24296      0 --:--:-- --:--:-- --:--:-- 24382
ubuntu@ip-172-31-41-179:/tmp$ less /tmp/script.deb.sh
ubuntu@ip-172-31-41-179:/tmp$
```

```
ubuntu@ip-172-31-41-179: /    ubuntu@ip-172-31-41-179: ~
  GNU nano 6.2                                  /etc/gitlab/gitlab.rb *
##! There are instances where this behavior is not possible or desired. For example, when providing passu
##! or connecting to third party services.
##! In those instances, we endeavour to provide an example configuration.

## GitLab URL
##! URL on which GitLab will be reachable.
##! For more details on configuring external_url see:
##! https://docs.gitlab.com/omnibus/settings/configuration.html#configuring-the-external-url-for-gitlab
##!
##! Note: During installation/upgrades, the value of the environment variable
##! EXTERNAL_URL will be used to populate/replace this value.
##! On AWS EC2 instances, we also attempt to fetch the public hostname/IP
##! address from AWS. For more details, see:
##! https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/instancedata-data-retrieval.html
external_url 'http://gitlab.vraj.pushkar.cloud'

## Roles for multi-instance GitLab
##! The default is to have no roles enabled, which results in GitLab running as an all-in-one instance.
##! Options:
##!   application_role
##!   redis_sentinel_role
##!   redis_master_role
##!   redis_replica_role
##!   monitoring_role
##!   geo_primary_role
##!   geo_secondary_role

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify   ^/ Go To Line M-E Redo
```

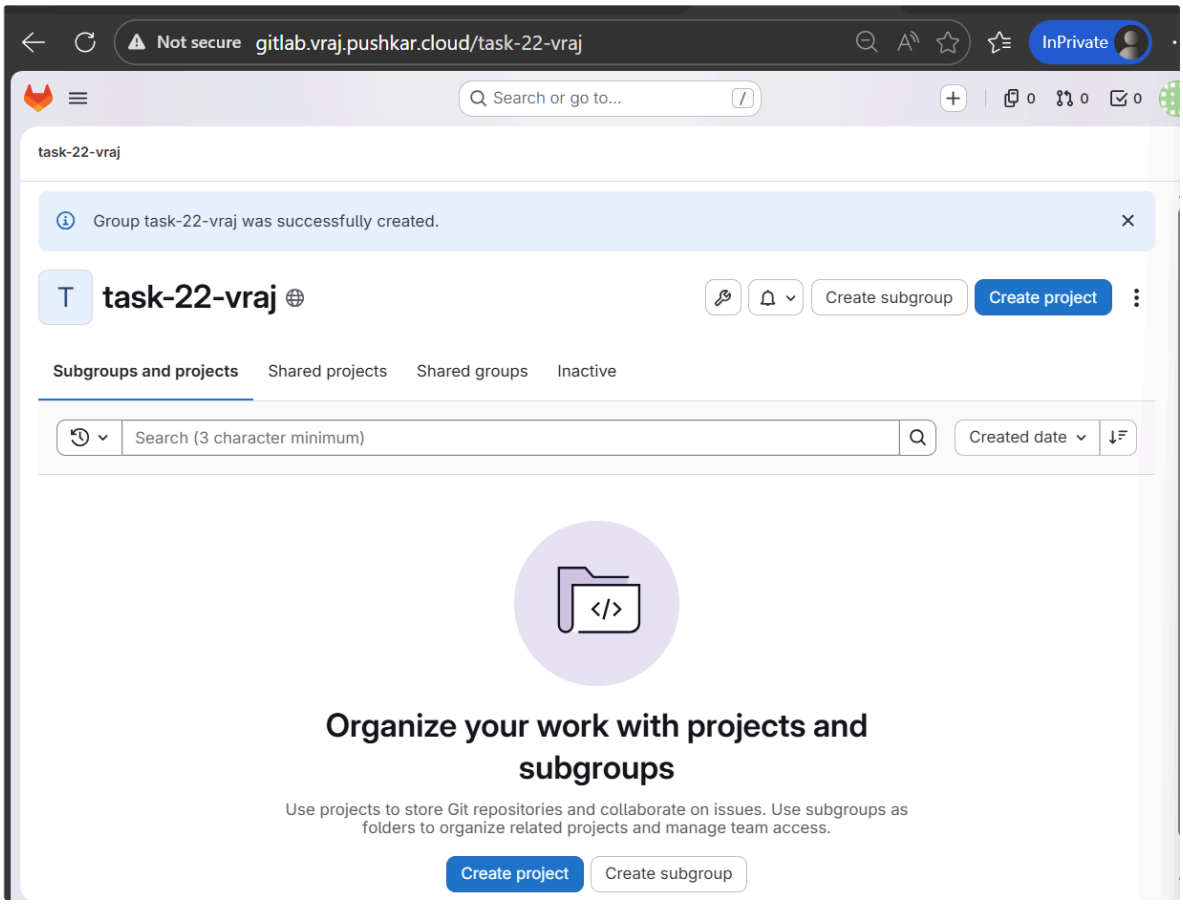
```
ubuntu@ip-172-31-41-179:/tmp$ sudo bash /tmp/script.deb.sh
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Get:5 https://packages.gitlab.com/gitlab/gitlab-ce/ubuntu/jammy jammy InRelease [16.9 kB]
Get:6 https://packages.gitlab.com/gitlab/gitlab-ce/ubuntu/jammy jammy/main amd64 Packages [30.3 kB]
Fetched 47.2 kB in 1s (75.9 kB/s)
Reading package lists... Done
Repository configured successfully.
Ready to install packages.
ubuntu@ip-172-31-41-179:/tmp$ cd ..
ubuntu@ip-172-31-41-179:/$ sudo apt install gitlab-ce
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  gitlab-ce
0 upgraded, 1 newly installed, 0 to remove and 79 not upgraded.
Need to get 1413 MB of archives.
After this operation, 4145 MB of additional disk space will be used.
Get:1 https://packages.gitlab.com/gitlab/gitlab-ce/ubuntu/jammy jammy/main amd64 gitlab-ce amd64 18.10.0-ce.0 [1413 MB]
Fetched 1413 MB in 19s (74.6 MB/s)
Selecting previously unselected package gitlab-ce.
(Reading database ... 66209 files and directories currently installed.)
Preparing to unpack .../gitlab-ce_18.10.0-ce.0_amd64.deb ...
Unpacking gitlab-ce (18.10.0-ce.0) ...

Progress: [ 20%] [#######################.....................................................................................]
```

### 4.3 Access GitLab Web Interface

GitLab was accessed using:

```
http://<EC2-Public-IP>
```

Initial root password was set and login was completed.

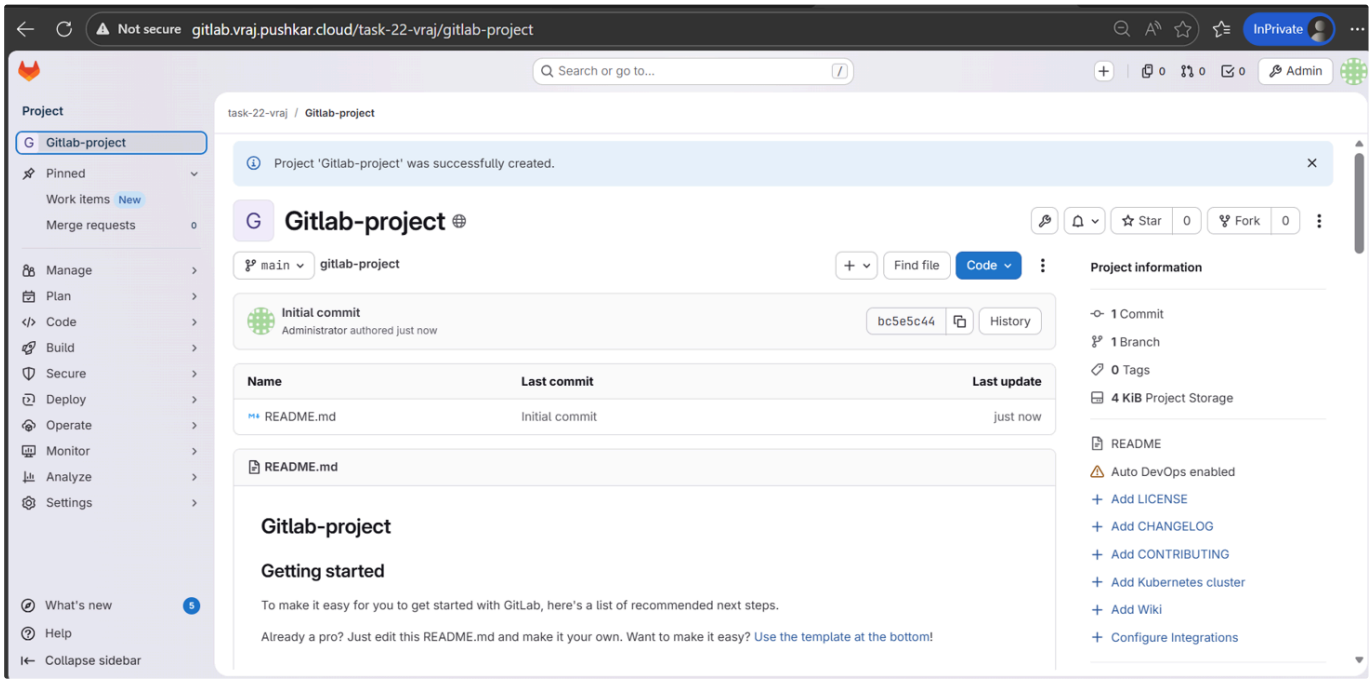


## 5. Repository Creation in GitLab

A new project/repository was created in GitLab for the Flask Hello World application.

Steps:

1. Login to GitLab
2. Click **New Project**
3. Enter Project Name
4. Set visibility
5. Create project

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## 6. Flask Hello World Application

### 6.1 Flask Application Code

Example Flask app:

```
from flask import Flask
app = Flask(__name__)

@app.route('/')
def home():
    return "Hello World from Flask App"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

Project structure:

```
flask-app/
│
├── app.py
├── requirements.txt
└── README.md
```

```
ubuntu@ip-172-31-41-179:/$ git clone https://github.com/pushkar-iamops/hello-world-flask.git
fatal: could not create work tree dir 'hello-world-flask': Permission denied
ubuntu@ip-172-31-41-179:/$ sudo su
root@ip-172-31-41-179:/# git clone https://github.com/pushkar-iamops/hello-world-flask.git
Cloning into 'hello-world-flask'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 4 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.
root@ip-172-31-41-179:/#
```

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## 7. Push Code to GitLab Repository

### 7.1 Initialize Git Repository

```
git init
git add .
git commit -m "Initial commit - Flask Hello World"
```

### 7.2 Add GitLab Remote Repository

```
git remote add origin http://<gitlab-ip>/<username>/<repo-name>.git
git branch -M main
git push -u origin main
```

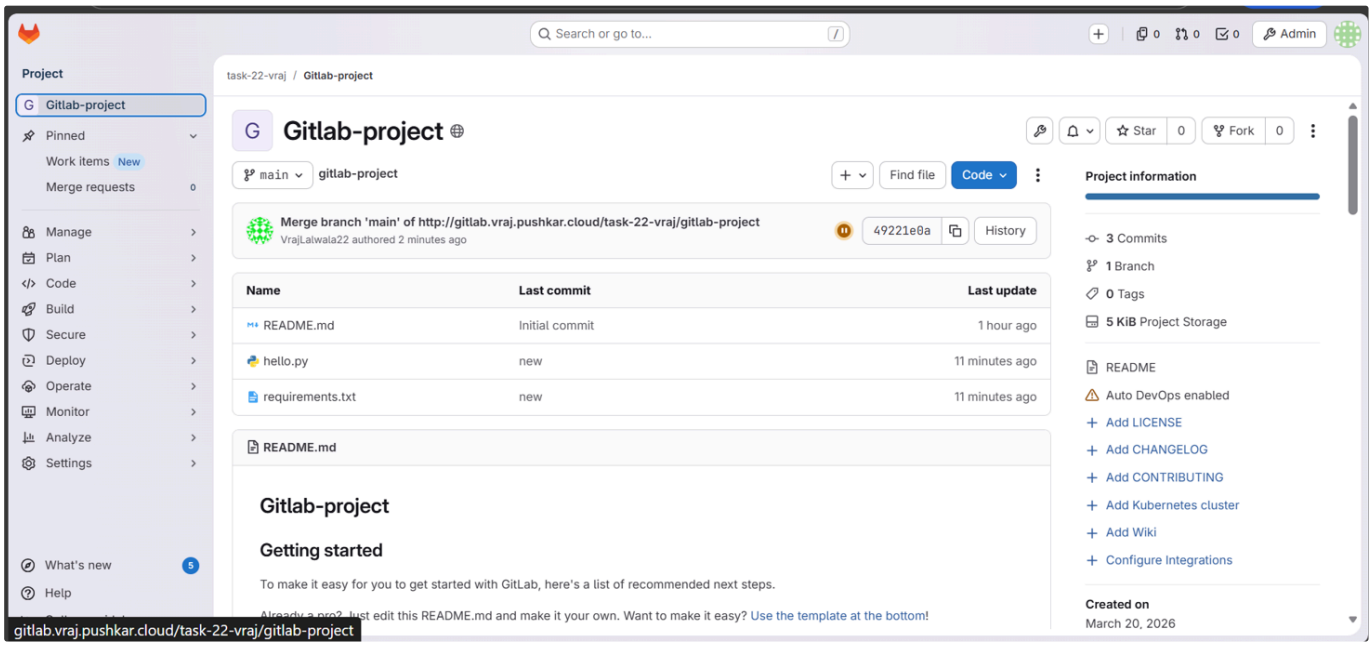
```
PS C:\Users\vraj.lalwala\OneDrive - Iamops\Desktop\Vraj-gitlab\hello-world-flask> git init
Initialized empty Git repository in C:/Users/vraj.lalwala/OneDrive - Iamops/Desktop/Vraj-gitlab/hello-world-flask/.git/
PS C:\Users\vraj.lalwala\OneDrive - Iamops\Desktop\Vraj-gitlab\hello-world-flask> git commit -m "new"
On branch master

Initial commit

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        hello.py
        requirements.txt

nothing added to commit but untracked files present (use "git add" to track)
PS> git add .                                                        world-flask>
PS C:\Users\vraj.lalwala\OneDrive - Iamops\Desktop\Vraj-gitlab\hello-world-flask> git add .
PS C:\Users\vraj.lalwala\OneDrive - Iamops\Desktop\Vraj-gitlab\hello-world-flask> git commit -m "new"
[master (root-commit) 62b0b63] new
 2 files changed, 17 insertions(+)
 create mode 100644 hello.py
 create mode 100644 requirements.txt
PS C:\Users\vraj.lalwala\OneDrive - Iamops\Desktop\Vraj-gitlab\hello-world-flask> git branch -M main
PS C:\Users\vraj.lalwala\OneDrive - Iamops\Desktop\Vraj-gitlab\hello-world-flask> git remote add origin http://gitlab.vraj.pushkar.cloud/task-22
-vraj/gitlab-project.git
PS C:\Users\vraj.lalwala\OneDrive - Iamops\Desktop\Vraj-gitlab\hello-world-flask> git push -uf origin main
fatal: Unencrypted HTTP is not recommended for GitLab. Ensure the repository remote URL is using HTTPS or see https://aka.ms/gcm/unsaferemotes a
bout how to allow unsafe remotes.
Username for 'http://gitlab.vraj.pushkar.cloud': root
Password for 'http://root@gitlab.vraj.pushkar.cloud':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 493 bytes | 493.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: GitLab: You are not allowed to force push code to a protected branch on this project.
To http://gitlab.vraj.pushkar.cloud/task-22-vraj/gitlab-project.git
```

```
PS C:\Users\vraj.lalwala\OneDrive - Iamops\Desktop\Vraj-gitlab\hello-world-flask> SET GCM_ALLOW_UNSAFE_REMOTES=true
PS C:\Users\vraj.lalwala\OneDrive - Iamops\Desktop\Vraj-gitlab\hello-world-flask> git pull origin main --allow-unrelated-histories
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 2.74 KiB | 155.00 KiB/s, done.
From http://gitlab.vraj.pushkar.cloud/task-22-vraj/gitlab-project
 * branch            main       -> FETCH_HEAD
 * [new branch]      main       -> origin/main
Merge made by the 'ort' strategy.
 README.md | 93 ++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
 1 file changed, 93 insertions(+)
 create mode 100644 README.md
PS C:\Users\vraj.lalwala\OneDrive - Iamops\Desktop\Vraj-gitlab\hello-world-flask> ls


    Directory: C:\Users\vraj.lalwala\OneDrive - Iamops\Desktop\Vraj-gitlab\hello-world-flask


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a---l         3/20/2026   1:56 PM            230 hello.py
-a---l         3/20/2026   3:25 PM           6156 README.md
-a---l         3/20/2026   1:56 PM             85 requirements.txt


PS C:\Users\vraj.lalwala\OneDrive - Iamops\Desktop\Vraj-gitlab\hello-world-flask> git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 850 bytes | 50.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
```

## 8. Validation

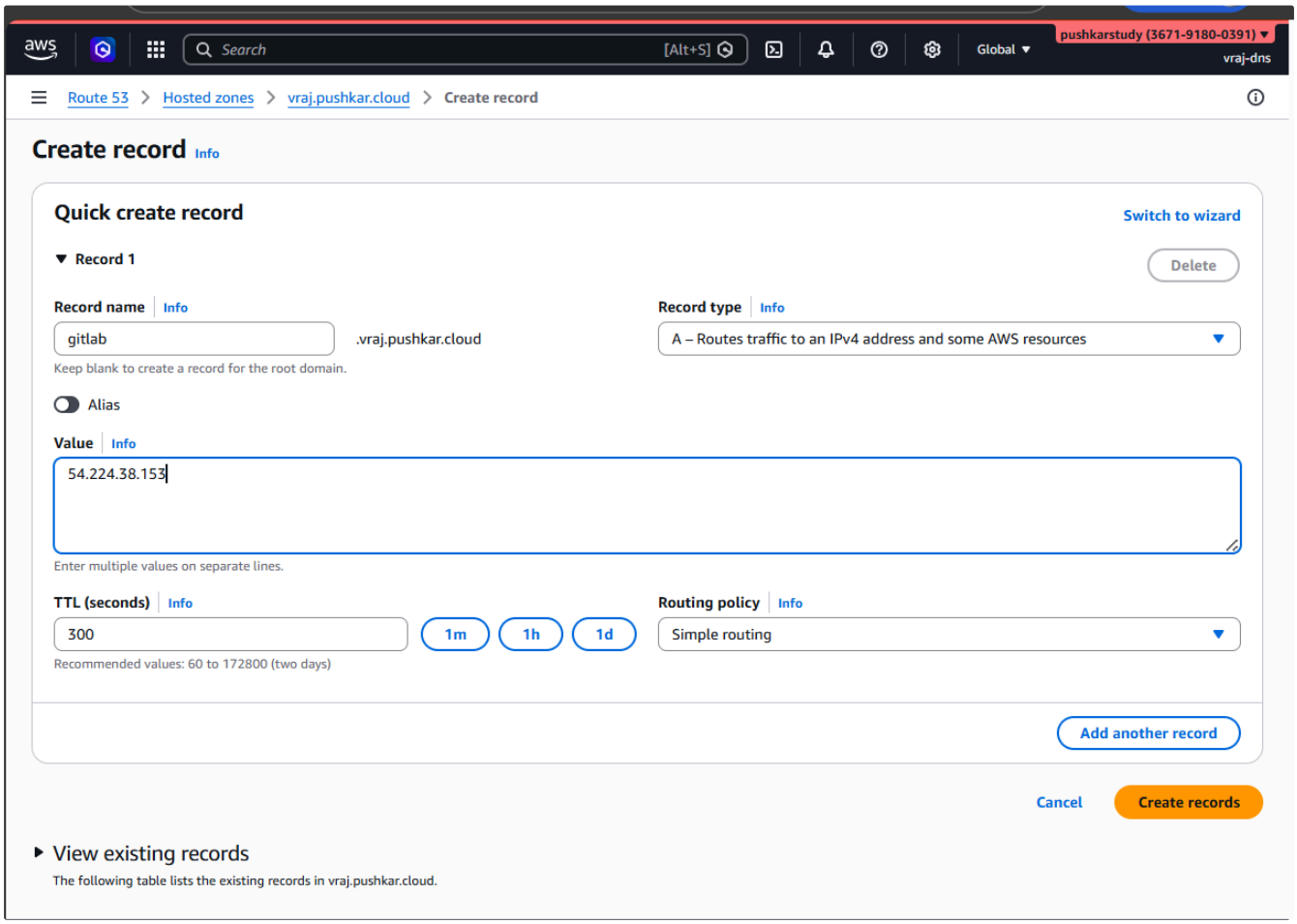
The following validations were performed:

| Validation | Status |
|---|---|
| GitLab Web Interface Accessible | Successful |
| Repository Created | Successful |
| Flask Code Uploaded | Successful |
| Git Push Working | Successful |



## 9. Security Considerations

- Only required ports were opened (22, 80, 443)
- SSH used for secure server access
- GitLab authentication enabled
- Repository access controlled via GitLab permissions



```
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##!   redis_master_role
##!   redis_replica_role
##!   monitoring_role
##!   geo_primary_role
##!   geo_secondary_role
##!   postgres_role
##!   patroni_role
##!   consul_role
##!   pgbouncer_role
```
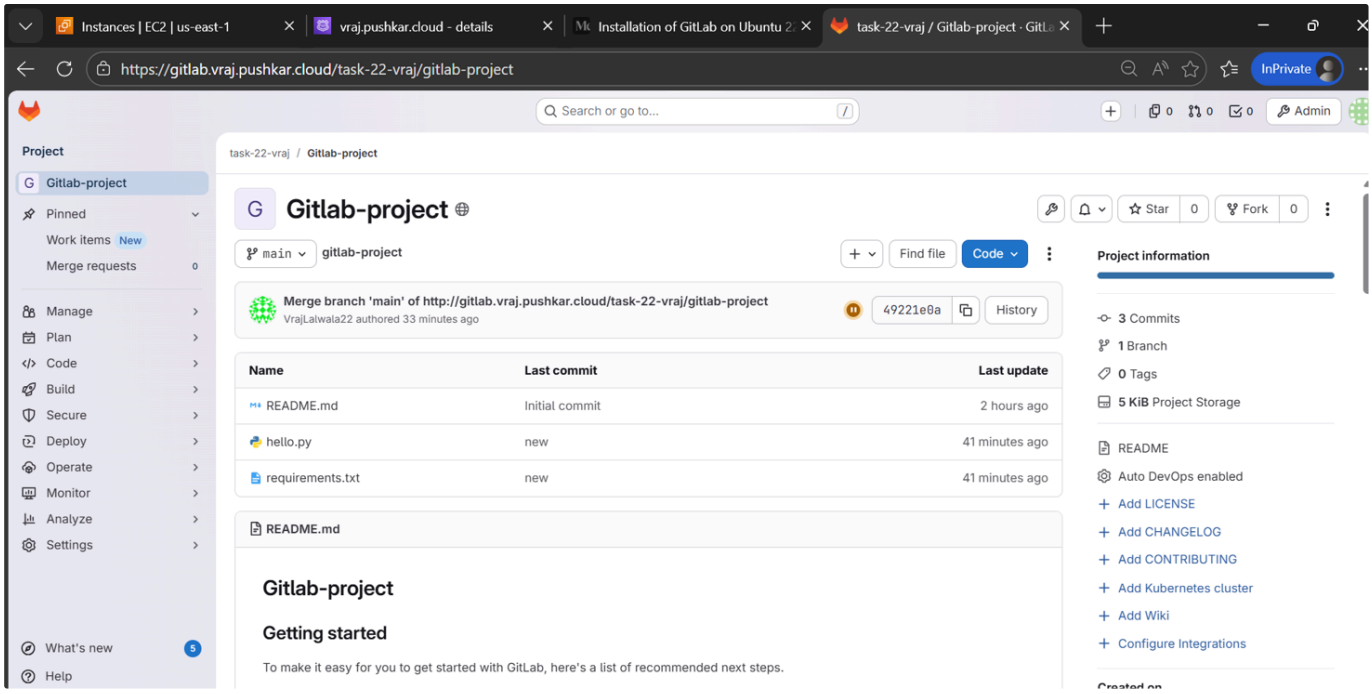

```
ubuntu@ip-172-31-41-179:/$ sudo nano /etc/gitlab/gitlab.rb
ubuntu@ip-172-31-41-179:/$ sudo gitlab-ctl reconfigure
[2026-03-20T10:24:26+00:00] INFO: Started Cinc Zero at chefzero://localhost:1 with repository at /opt/gitlab/embedded (One version pe
r cookbook)
Cinc Client, version 18.3.0
Patents: https://www.chef.io/patents
Infra Phase starting
[2026-03-20T10:24:26+00:00] INFO: *** Cinc Client 18.3.0 ***
[2026-03-20T10:24:26+00:00] INFO: Platform: x86_64-linux
[2026-03-20T10:24:26+00:00] INFO: Cinc-client pid: 31262
[2026-03-20T10:24:26+00:00] INFO: Setting the run_list to ["recipe[gitlab-base]"] from CLI options
[2026-03-20T10:24:26+00:00] INFO: Run List is [recipe[gitlab-base]]
[2026-03-20T10:24:26+00:00] INFO: Run List expands to [gitlab-base]
[2026-03-20T10:24:26+00:00] INFO: Starting Cinc Client Run for ip-172-31-41-179.ec2.internal
[2026-03-20T10:24:26+00:00] INFO: Running start handlers
[2026-03-20T10:24:26+00:00] INFO: Start handlers complete.
Resolving cookbooks for run list: ["gitlab-base"]
[2026-03-20T10:24:27+00:00] INFO: Loading cookbooks [gitlab-base@0.1.0, package@0.1.0, gitlab@0.0.1, runit@5.1.7, logrotate@0.1.0, po
stgresql@0.1.0, redis@0.1.0, monitoring@0.1.0, registry@0.1.0, mattermost@0.1.0, consul@0.1.0, gitaly@0.1.0, praefect@0.1.0, gitlab-k
as@0.1.0, gitlab-pages@0.1.0, letsencrypt@0.1.0, nginx@0.1.0, acme@4.2.2, crond@0.1.0]
Synchronizing cookbooks:
  - gitlab-base (0.1.0)
```




## 10. Challenges Faced

| Issue | Solution |
|---|---|
| GitLab installation took long time | Increased EC2 instance size |
| Git push authentication issue | Used correct GitLab credentials |
| Port access issue | Updated security group |

## 11. Difference Between GitLab, GitHub, and Jenkins

| Feature | GitLab | GitHub | Jenkins |
|---|---|---|---|
| Type | DevOps Platform | Code Hosting | CI/CD Tool |
| Repository | Yes | Yes | No |
| CI/CD | Built-in | GitHub Actions | Yes |
| Hosting | Self-hosted | Cloud | Self-hosted |
| Use | Full DevOps | Code Hosting | Automation |

**Simple Explanation:**

- GitHub → Code hosting
- GitLab → Code hosting + CI/CD
- Jenkins → Automation tool for CI/CD

## 12. Conclusion

This project successfully demonstrated the installation and configuration of a self-hosted GitLab CE server on AWS EC2. A Flask Hello World application repository was created and code was successfully pushed to GitLab. This setup provides an internal version control system for managing source code and supports collaboration and version tracking.